METRIC
MIL-S-24795(SH)
22 June 1995

MILITARY SPECIFICATION

SENSORS, TEMPERATURE, FIBER OPTIC (NAVAL SHIPBOARD USE)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single-channel, fiber optic temperature sensors (FOTS) for Naval shipboard use, but does not include the readout display.

1.2 Classification (see 6.2).

1.2.1 Designations. The FOTS classification shall consist of a series of designations which shall be assigned and listed in the format below.

24795	-	XX	-	XX	-	X	-	X	-	X
Specification		Type		Input Power		Opto-Electronics Module		Signal Output		Range
		1.2.1.1		1.2.1.2		1.2.1.3		1.2.1.4		1.2.1.5

1.2.1.1 Type. The sensor head type shall be designated as follows:

T1 - Thermowell (see table I)
T2 - Thermowell (see table I)
T3 - Thermowell (see table I)
T4 - Thermowell (see table I)
EM - Embedded (Bearing Applications, see table II)
B1 - Bare-bulb (see table III)
B2 - Bare-bulb (see table III)

1.2.1.2 Input power. The input power required to operate the opto-electronics module shall be designated as follows:

DC - 28 Volts direct current (Vdc)
AC - 115 Volts alternating current (Vac)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6040

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1.2.1.3 Opto-electronics module. The opto-electronics module type shall be designated as follows:

A - Junction box enclosure
B - Control console module
C - Monitoring and alarm panel module

1.2.1.4 Signal output. The signal output shall be designated as follows:

A - 4 to 20 mA direct current
V - 0 to 10 volts direct current
O - Digital optical output
E - Digital electrical output

1.2.1.5 Range. The FOTS measurement range shall be designated as follows:

1 - -40 to 150°C
2 - 0 to 300°C
3 - -40 to 300°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks listed in that issue of Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	-	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MS3452	-	Connector, Receptacle, Electric, Box Mounting, Rear Release, Crimp Contact, AN Type with.
MS3456	-	Connector, Plug, Electrical, Rear Release, Crimp Contact, AN Type.
MIL-C-5015	-	Connector, Electrical, Circular Threaded, AN Type, General Specification for.
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts): Packaging of.
MIL-S-19622	-	Stuffing Tubes, Nylon; and Packing Assemblies; General Specification for.
MIL-S-19622/1	-	Stuffing Tubes, Straight, Nylon.

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MIL-S-19622/17	-	Stuffing Tube, Nylon, Size 2: Packing Assemblies for.
MIL-E-24142	-	Enclosures for Electrical Fittings and Fixtures, General Specification for.
MIL-E-24142/3	-	Enclosure for Electrical Fittings and Fixtures, Submersible, Size 6 by 9 (15 Foot).
MIL-T-24270	-	Thermowells for Thermometers and Electrical Temperature Sensors, General Specification for.
MIL-M-24794	-	Material, Index Matching, Fiber Optics.
MIL-F-49291	-	Fiber, Optical, (Metric), General Specification for.
MIL-P-83461	-	Packings, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C).
MIL-P-83461/1	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C) Sizes and Tolerances.
MIL-C-83522	-	Connectors, Fiber Optic, Single Terminus, General Specification for.
MIL-C-83522/16	-	Connector, Fiber Optic, Single Terminus, Plug, Adapter Style, 2.5 Millimeter Bayonet Coupling, Epoxy.
MIL-C-83522/17	-	Connector, Fiber Optic, Single Terminus, Adapter, 2.5 Millimeter Bayonet Coupling, Bulkhead Panel Mount.
MIL-C-83522/18	-	Connector, Fiber Optic, Single Terminus, Adapter, 2.5 Millimeter Bayonet Coupling, PC Mount.
MIL-C-85045	-	Cables, Fiber Optic, (Metric), General Specification for.

STANDARDS

MILITARY

MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Requirements for Control of Electromagnetic Interference Emission and Susceptibility.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.

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MILITARY - Continued

MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1861	-	Electrical and Electronic Assemblies, Boards, Cards, and Associated Hardware, Selection and Use of.
MIL-STD-2042	-	Fiber Optic Topology Installation Standard Methods for Naval Ships.

HANDBOOKS

MILITARY

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the non-Government documents which are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 23	-	Standard Specification for White Metal Bearing Alloys Known Commercially as "Babbitt Metal".
D 3951	-	Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-422	-	Electrical Characteristics of Balanced Voltage Digital Interface Circuit.

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein , the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Reliability. A reliability prediction for the electronic and active opto-electronic components (see 3.4.2), in the form of a mean time between failure (MTBF) calculation, shall be performed. The computational techniques specified in MIL-HDBK-217 should be used for guidance. The calculated MTBF shall be no less than 40,000 hours.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.5).

3.3 Materials.

3.3.1 Metals. Unless otherwise specified herein, all metals used in the construction of the FOTS shall be corrosion-resistant and shall be in accordance with MIL-STD-454, requirement 15. Dissimilar metals shall not be used in intimate contact with each other unless suitably finished to prevent electrolytic corrosion and shall be in accordance with MIL-STD-454, requirement 16. Materials used to fabricate the sensor head for the embedded configuration shall have a Brinell hardness less than 170 to ensure that shaft scoring does not occur.

3.3.2 Flammable materials. Materials used in the construction of the FOTS shall, in the end configuration, be noncombustible or fire retardant in the most hazardous conditions of atmosphere, pressure, and temperature to be expected in the application. Fire retardant additives may be used provided they do not adversely affect the specified performance requirements of the basic materials. Fire retardance shall not be achieved by use of nonpermanent additives to the basic material.

3.3.3 Fungus resistant materials. Materials used in the construction of the FOTS shall be in accordance with MIL-STD-454, requirement 4 for fungus inert materials. Materials that are not in accordance with requirement 4 of MIL-STD-454 shall be in accordance with grade I classification of MIL-STD-810, method 508.

3.3.4 Solvents, adhesives, and cleaning agents. If any chemicals or cements are used in bonding of internal components, no degradation shall result during in-service use. MIL-STD-454, requirement 23 may be used as guidance for epoxy and adhesives used in construction of the FOTS.

3.3.5 Refractive index matching gels, fluids, or compounds. Refractive index matching gels, fluids, or compounds shall be in accordance with MIL-M-24794.

3.4 Design and construction. FOTS shall convert the thermal energy surrounding the sensor head to a continuous linear analog dc signal or specified digital output throughout a specified measurement range. FOTS shall consist of a sensor head, opto-electronics module, and fiber optic cable

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interconnect. All parts of the FOTS shall be interchangeable with the appropriate replacement parts with respect to form, fit, and function; and maintain the required accuracy specified in 3.5.1.

3.4.1 Sensor head. The sensor head shall detect the surrounding temperature. The sensor head shall be passive in nature by detection of temperature through a change in optical properties. Neither electrical nor electronic components shall be used in the construction of the sensor head.

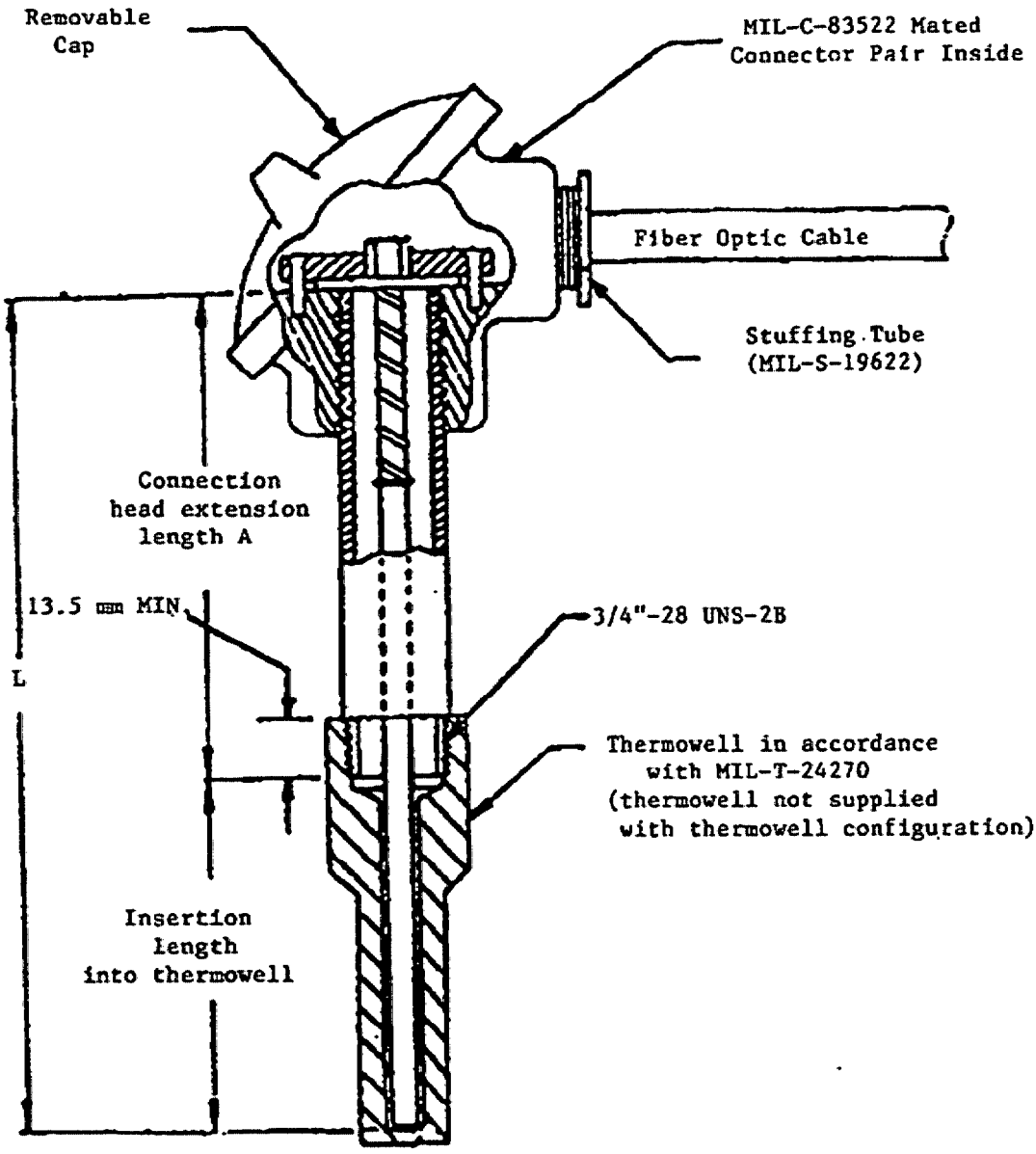
3.4.1.1 Thermowell type. The sensor head shall be constructed in accordance with figures 1 and 2. The sensor head dimensions shall be in accordance with table I. The sensor head shall have a weight not greater than one kilogram.

TABLE I. Thermowell type sensor head dimensions.

Thermowell type sensor dimensions in mm(in.); See figures 1 and 2						
Type	Insertion length into thermowell	L ± 1.02 (0.04)	A ± 0.76 (0.03)	Y ± 0.25 (0.01)	Z ± 4.06 (0.16)	MIL-T-24270 thermowell type applicable
T1	68.83 (2.71)	113.28 (4.46)	44.45 (1.75)	95.25 (3.75)	133.35 (5.25)	2C
T2	68.83 (2.71)	246.63 (9.71)	177.80 (7.00)	228.60 (9.00)	266.70 (10.5)	2C
T3	119.63 (4.71)	164.08 (6.46)	44.45 (1.75)	146.05 (5.75)	184.15 (7.25)	4C
T4	119.63 (4.71)	259.33 (10.21)	139.70 (5.50)	241.30 (9.50)	279.40 (11.00)	4C

NOTE: The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

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3.4.1.
2
Embedd

FIGURE 1. Thermowell type sensor head assembly.

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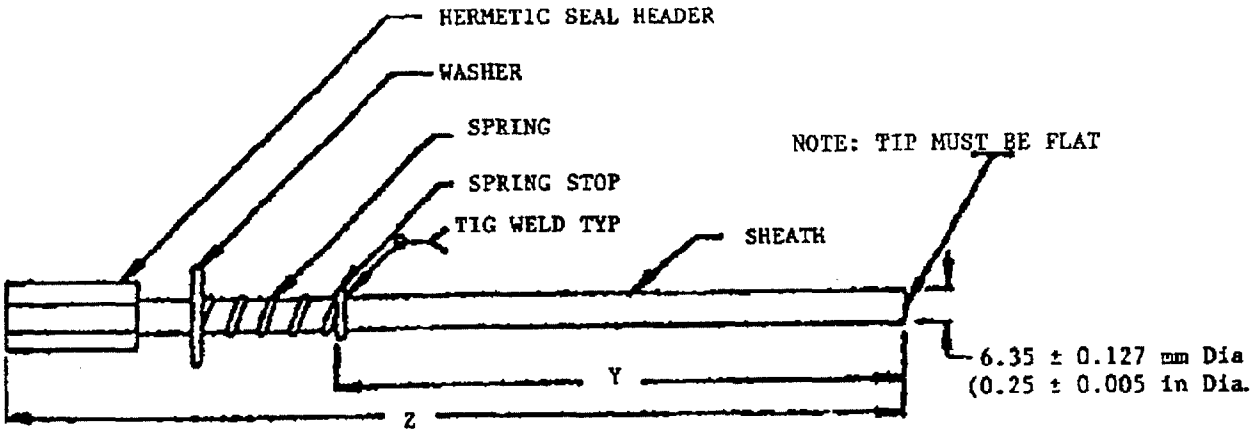


FIGURE 2. Sensor head sheath construction

ed type. The sensor head shall be constructed in accordance with the dimensions shown on figure 3 and in table II. The babbitt topping material shall be in accordance with ASTM B 23 (grade 2).

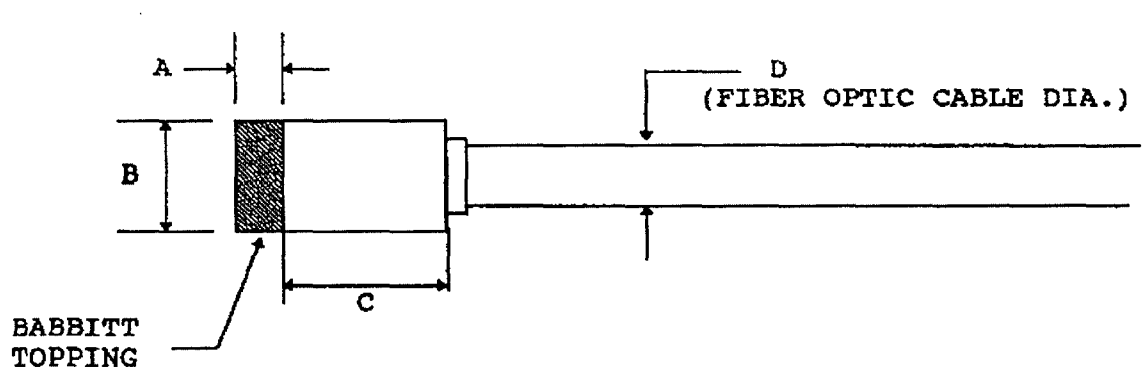


FIGURE 3. Embedded type sensor head assembly.

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TABLE II. Embedded type sensor head dimensions.

Embedded type sensor head dimensions in mm(in.); See figure 3	
A	2.54 +1.52/-0.00 (0.10 +0.06/-0.00)
B	6.96 ± 0.05 (0.274 ± 0.002)
C	6.35 +0.00/-0.13 (0.250 +0.00/-0.005)
D	3.18 ± 0.41 (0.125 ± 0.16)

3.4.1.3 Bare-bulb type. The sensor head dimensions shall be in accordance with figure 4 and table III. The sensor head shall have a weight not greater than 1 kilogram. The sensor sheath shall be designed to withstand application conditions of 37.66 MPa (5500 pounds per square inch), 210°C, and an air flow velocity of 12.2 meters (40 feet) per second. Conformance shall be proven by using the mathematical model from the technical paper entitled "Power Test Code Thermometer Wells" (J. W. Murdock, Journal of Engineering for Power, ASME, October 1959, pp 403-416).

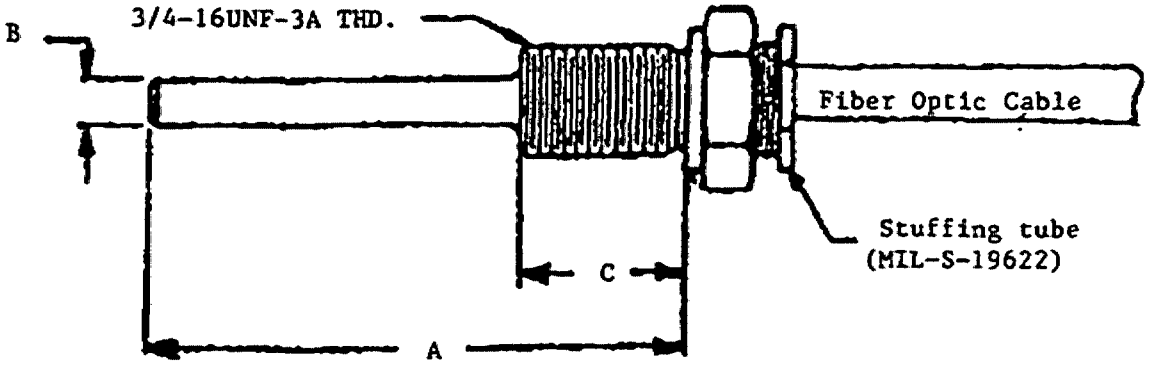


FIGURE 4. Bare-bulb sensor head assembly.

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TABLE III. Bare-bulb type sensor head dimensions.

Bare-bulb type sensor head dimensions in mm(in.); See figure 4			
	A	B	C (minimum)
Type B1	5.72 ± 0.76 (2.25 ± 0.03)	9.53 +0.13/-0.00 (0.375 +0.005/-0.00)	22.2 (0.875)
Type B2	16.51 ± 0.76 (6.50 ± 0.03)	9.53 +0.13/-0.00 (0.375 +0.005/-0.00)	22.2 (0.875)

NOTE: The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.4.2 Opto-electronics module.

3.4.2.1 Junction box enclosure (Type A). The opto-electronics shall be housed in a junction box in accordance with MIL-E-24142 and MIL-E-24142/3. The opto-electronics module shall have a weight not greater than 4.5 kilograms.

3.4.2.2 Control console module (Type B). The opto-electronics shall be packaged in a circuit card which is a modular subassembly of a control console. Design and test requirements for the opto-electronics module shall be as specified (see 6.2).

3.4.2.3 Monitoring and alarm panel module (Type C). The opto-electronics shall be packaged in a module which is a subassembly of a bulkhead mounted, temperature monitoring and alarm panel. Design and test requirements for the opto-electronics module shall be as specified (see 6.2).

3.4.3 Fiber optic cable.

3.4.3.1 Thermowell and bare-bulb sensor types. The cable used to interface the sensor head and the opto-electronics module shall have an outer diameter of a four fiber cable in accordance with MIL-C-85045. In the cable there shall be no less than two times the number of fibers required for operation of the sensor. The cable shall be supplied with a stuffing tube (size 2, military part number M19622/1-002 in accordance with MIL-S-19662/1) and packing assembly (military part number M19622-17-0001 in accordance with MIL-S-19662/17) in accordance with MIL-S-19622, and an O-ring (military part number M83461/1-214) in accordance with MIL-P-83461/1 and MIL-P-83461 installed on each end of the cable to accomplish watertight penetration into the sensor head and opto-electronic module. Exposed single fiber optical fiber cable component shall not be used over distances greater than one meter. The cable length shall be as specified (see 6.2).

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3.4.3.2 Embedded sensor type. The cable used to interface the sensor head and a local interconnection box shall be in accordance with MIL-C-85045. The cable length shall be as specified (see 6.2).

3.4.4 Optical fiber. Optical fiber used to transmit light between the opto-electronics module and the sensor head shall be in accordance with MIL-F-49291.

3.4.5 Fiber optic connectors, receptacles, and bulkhead adapters. All fiber optic connectors and, receptacles, and bulkhead adapters shall be in accordance with MIL-C-83522 and MIL-C-83522/16, 17, and 18 respectively. The fiber optic cable for the thermowell sensor type shall be terminated at both ends with connectors. The fiber optic cable for the embedded and bare-bulb sensor configurations shall be terminated with a connector at the end which interfaces with the opto-electronic module. MIL-STD-2042 method 5B1 may be used as guidance. The connectors and receptacles shall be mounted inside the sensor head (thermowell sensor type only) and the opto-electronic module.

3.4.6 Electrical input power. Nominal steady-state power supply requirements for ac shall be 115 ± 8 Vac, 60 ± 2 Hertz, single phase. Nominal steady-state power supply requirements for dc shall be 28 ± 4.5 Vdc. The sensor shall operate with power supply variations as specified (see 4.7.7 and 4.7.8).

3.4.7 Output signal.

3.4.7.1 Current output. When a current output is required, the output signal shall be directly proportional to the temperature being measured. The 4 mA output shall correspond to the lower temperature range value and the 20 mA output shall correspond to the upper temperature range value. The current output shall remain accurate regardless of external load resistance variations over a range of 0 to 250 ohms.

3.4.7.2 Voltage output. When a voltage output is required, the output signal shall be directly proportional to the temperature being measured. The 0 Vdc output shall correspond to the lower temperature range value and the 10 Vdc output shall correspond to the upper temperature range value. The voltage output shall remain accurate regardless of external load resistances greater than 100 thousand ohms.

3.4.7.3 Digital optical output. When a digital optical output is required, the output signal specifications shall be as specified (see 6.2).

3.4.7.4 Digital electrical output. When a digital electrical output is required, all requirements shall be as specified (see 6.2). The electrical characteristics shall be those specified in EIA standards RS-422 for balanced voltage digital interface circuitry, or as specified (see 6.2). The data format shall be as specified (see 6.2).

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3.4.8 Electrical connectors. A single electrical receptacle in accordance with MIL-C-5015 shall be used to interface the input power and linear output signal to the opto-electronics module. When a digital optical or digital electrical output is specified (see 1.2.1.4), connectors shall be as specified (see 6.2). The appropriate connector assembly and pin designations for each of the possible FOTS configurations shall be as specified in 3.4.8.1 and 3.4.8.2.

3.4.8.1 DC input power. The receptacle mounted to the opto-electronics module shall be classification MS3452W14S-5PX in accordance with MS3452. Receptacle pin "A" shall be positive 28 Vdc input power, pin "B" shall be negative input power, pin "C" shall be case ground, pin "D" shall be positive mA or high Vdc output signal, and pin "E" shall be negative mA or low Vdc output signal. The mating plug supplied with the FOTS shall be classification MS3456W14S-5SX in accordance with MS3456.

3.4.8.2 AC input power. The receptacle mounted to the opto-electronics module shall be classification MS3452W14S-5PX in accordance with MS3452. Receptacle pin "A" shall be positive 115 Vac input power, pin "B" shall be input power return, pin "C" shall be case ground, pin "D" shall be positive mA or high Vdc output signal, and pin "E" shall be negative mA or low Vdc output signal. The mating plug supplied with the FOTS shall be classification MS3456W14S-5SX in accordance with MS3456.

3.4.9 Adjustments. A means shall be provided for adjusting the output signal of the opto-electronics module for calibration purposes. The adjustments shall be situated in the opto-electronics module so that calibrations can be performed while the FOTS is energized.

3.4.10 Printed circuit boards. Printed circuit boards shall be in accordance with MIL-STD-1861.

3.4.11 Fuses. The opto-electronics module shall not be fused.

3.5 Performance requirements.

3.5.1 Accuracy. The FOTS output signal accuracy shall be within plus or minus 1 percent of span (see 4.7.1). Span is defined as the algebraic difference between the limits of the measurement range.

3.5.2 Repeatability. The FOTS output signal at each temperature point shall not change in value more than 0.5 percent of span during the repeatability test (see 4.7.2).

3.5.3 Reference measurement. The FOTS output signal accuracy shall be within plus or minus 1 percent of span (see 4.7.3).

3.5.4 Sensitivity factor. The ratio of the FOTS output signal percentage change of span to applied bath temperature percentage change of span shall not be less than 0.75 nor greater than 1.25 (see 4.7.4).

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3.5.5 Response time. Response time of the FOTS output signal shall be not greater than 8 seconds for the thermowell sensor type, not greater than 5 seconds for the embedded sensor type, and not greater than 15 seconds for the bare-bulb sensor type (see 4.7.5).

3.5.6 Warm-up time. The FOTS output signal accuracy at the bath temperature shall be within plus or minus 1 percent of span within 1 minute after the FOTS is energized (see 4.7.6).

3.5.7 Steady-state supply voltage and frequency (ac) or supply voltage (dc). The FOTS output signal shall be within plus or minus 1 percent of span throughout the test (see 4.7.7).

3.5.8 Transient voltage and frequency (ac) or voltage (dc). The FOTS output signal accuracy shall be within plus or minus 1 percent of span (see 4.7.8).

3.5.9 Insulation resistance. The insulation resistance of the sensor shall not be less than 10 megohms (see 4.7.9).

3.5.10 Power interruption. The FOTS output signal accuracy shall be within plus or minus 1 percent of span (see 4.7.10).

3.5.11 Short circuit. The FOTS output signal accuracy shall be within plus or minus 1 percent of span (see 4.7.11).

3.5.12 Line voltage reversal (dc). The FOTS output signal accuracy shall be within plus or minus 1 percent of span (see 4.7.12).

3.5.13 Temperature. The FOTS shall operate without interruption throughout the test. The FOTS output signal accuracy shall be within plus or minus 1 percent of span at the specified test measurement points (see 4.7.13).

3.5.14 Humidity. There shall be no evidence of water penetration into the FOTS sensor head either during or at the conclusion of the humidity test. The FOTS output signal shall be within plus or minus 1 percent of span at the conclusion of the test (see 4.7.14).

3.5.15 Enclosure. There shall be no evidence of water penetration into the FOTS sensor head either during or at the conclusion of the enclosure test. The FOTS output signal shall be within plus or minus 1 percent of span at the conclusion of the test (see 4.7.15).

3.5.16 Salt spray. No corrosion or other damage shall be evident at the conclusion of the salt spray test. The FOTS output signal shall be within plus or minus 1 percent of span at the conclusion of the test (see 4.7.16).

3.5.17 Overtemperature. The sensor head shall show no evidence of physical damage at the conclusion of the test and the reference measurement results shall conform to 3.5.3 (see 4.7.17).

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3.5.18 Accelerated life. The FOTS shall operate without interruption throughout the test. The sensor head shall show no evidence of physical damage at the conclusion of the test and the reference measurement results shall conform to 3.5.3 (see 4.7.18).

3.5.19 Vibration. The FOTS shall operate without interruption and show no evidence of physical damage throughout the vibration test. The reference measurement results shall conform to 3.5.3 at the conclusion of the test (see 4.7.19).

3.5.20 Shock. The FOTS shall operate without interruption and show no evidence of physical damage throughout the shock test. The reference measurement results shall conform to 3.5.3 at the conclusion of the test (see 4.7.20).

3.5.21 Electromagnetic interference (EMI) emission and susceptibility. The FOTS shall operate without interruption and shall be in accordance with the requirements of MIL-STD-461: CE101, CE102, CS101, CS114, CS116, RE101, RE102, RS101, RS103 and RS105 (see 4.7.21 and 6.2).

3.5.22 Pressure. The FOTS shall operate without interruption during the test. The reference measurement results shall conform to 3.5.3 (see 4.7.22).

3.6 Cleaning and surface finishes. Surfaces of castings, forgings, molded parts, stampings, machined and welded parts shall be free of defects such as cracks, porosity, undercuts, voids, and gaps as well as sand, dirt, fins, sharp edges, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and edges shall be either rounded or beveled. There shall be no burn-through. There shall be no warpage or dimensional change due to heat from welding operation. There shall also be no damage to adjacent parts resulting from welding.

3.7 Identification marking.

3.7.1 Opto-electronics module. Each opto-electronics module shall be permanently and legibly marked in accordance with MIL-STD-130. The following minimum information shall be provided: nomenclature, FOTS classification (see 1.2), national stock number, manufacturer's name and model number, temperature range, contract number, and a unique serial number from the manufacturer.

3.7.2 Sensor head. Each sensor head shall be permanently and legibly marked in accordance with MIL-STD-130. The following minimum information shall be provided: nomenclature, FOTS classification (see 1.2), manufacturer's name and model number, and a unique serial number from the manufacturer. The embedded sensor type shall use a removable metallic tag which attaches to the cable interface for the identification marking. The following notice shall be marked on the opposite side of the tag:

> "NOTICE: Reattach this tag to the cable interface after bearing installation and keep affixed after the bearing is placed in service."

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3.8 Warning labels.

3.8.1 Alternating current. FOTS opto-electronics modules with input power designation AC (see 1.2.1.2) shall be permanently and legibly marked in accordance with MIL-STD-130. The following label shall be provided:

"WARNING 115 VAC"

3.8.2 Labeling. A visible label shall be affixed to the outside of the opto-electronics module cover and shall contain the following.

NOTICE
UNTERMINATED OPTICAL CONNECTIONS MAY
EMIT LASER RADIATION
DO NOT VIEW BEAM WITH OPTICAL INSTRUMENTS
AND AVOID DIRECT EXPOSURE TO THE BEAM

A visible label shall be affixed to the sensor head and the inside of the opto-electronics module and shall contain the following.

"WARNING
INVISIBLE LASER RADIATION
AVOID EXPOSURE TO THE BEAM"

The labels shall be yellow lettering on a black background.

3.9 Workmanship. Workmanship of the sensors shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specifications shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specifications shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance

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to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Standard operating conditions shall be established for conducting the test in this specification and shall only be changed when they are the variables in the tests.

4.4 First article inspection. When a first article inspection is required (see 3.2 and 6.2), samples shall be inspected as specified in table IV. Failure of any FOTS to meet the requirements of this specification shall be cause for rejection.

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TABLE IV. First article inspection.

Examination or test	Requirement	Procedure
Group I: One sample required		
General examination	3.3, 3.4, 3.6-3.9	4.6
Accuracy	3.5.1	4.7.1
Repeatability	3.5.2	4.7.2
Sensitivity	3.5.4	4.7.4
Response time	3.5.5	4.7.5
Warm-up time	3.5.6	4.7.6
Steady-state voltage and frequency	3.5.7	4.7.7
Transient voltage and frequency	3.5.8	4.7.8
Insulation resistance	3.5.9	4.7.9
Power interruption	3.5.10	4.7.10
Short circuit	3.5.11	4.7.11
Line voltage reversal (dc)	3.5.12	4.7.12
Temperature	3.5.13	4.7.13
Humidity	3.5.14	4.7.14
Enclosure	3.5.15	4.7.15
Salt spray	3.5.16	4.7.16
Overtemperature	3.5.17	4.7.17
Accelerated life	3.5.18	4.7.18
Vibration	3.5.19	4.7.19
Shock	3.5.20	4.7.20
Group II: One sample required		
General examination	3.3, 3.4, 3.6-3.9	4.6
Accuracy	3.5.1	4.7.1
Repeatability	3.5.2	4.7.2
EMI	3.5.21	4.7.21
Group IV: One sample required		
Pressure	3.5.22	4.7.22

4.4.1 Sample size. Two FOTS of each lot shall be subjected to the first article inspection. One sample shall be subjected to the test of Group I and one sample shall be subjected to the tests of Group II. Each sample shall be supplied with the length of cable required for the intended application, but shall be not less than 30.5 meters (100 feet). The Group IV test applies only to the embedded sensor type. The test in Group IV requires an additional embedded sensor due to the destructive nature of the test.

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4.4.2 Inspection lot. An inspection lot shall consist of all FOTS of the same classification (see 1.2), produced under essentially the same conditions, in the same facility from the same materials and offered for delivery at the same time.

4.4.3 Order of inspection. The sample FOTS shall be subjected to the inspections specified in table IV in the order listed. Any deviation in the test order must first be approved by NAVSEA.

4.4.4 Disposition of first article units. Samples subjected to first article inspection shall be considered consumed and non-deliverable as part of the contract. Final disposition of first article samples shall be as specified (see 6.2).

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests as specified in table V.

TABLE V. Quality conformance inspection.

Examination or test	Requirement	Procedure
Group A (All FOTS)		
General examination	3.3, 3.4, 3.6-3.9	4.6
Accuracy	3.5.1	4.7.1
Repeatability	3.5.2	4.7.2
Group B		
Steady-state voltage and frequency	3.5.9	4.7.9
Insulation resistance	3.5.11	4.7.11
Temperature	3.5.15	4.7.15
Enclosure	3.5.17	4.7.17

4.5.1 Inspection lot. An inspection lot shall consist of all FOTS of the same classification (see 1.2), produced under essentially the same conditions, in the same facility from the same materials and offered for delivery at the same time.

4.5.2 Order of inspection. The sample FOTS shall be subjected to the inspections specified in table V in the order listed. Any deviation in the test order must first be approved by NAVSEA.

4.5.3 Group A inspection. Each FOTS in each lot offered for delivery shall be subjected to the group A inspection specified in table V in the order listed. Failure of any FOTS to meet the requirements of this specification shall be cause for rejection.

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4.5.4 Group B inspection. Sample FOTS shall be selected in accordance with table VI. Each sample selected shall be subjected to the group B inspection specified in table V. Failure of any FOTS to meet the requirements of this specification shall be cause for rejection.

TABLE VI. Group B inspection sample size.

Lot or batch size	Sample size	Lot or batch size	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1200	80
26 to 50	8	1201 to 4200	125
51 to 90	13	4201+	3 per 100
91 to 150	20		

4.6 General examination. Each FOTS shall be given a thorough examination to determine conformance to the requirements of this specification with respect to material, finish, workmanship, construction, assembly, dimensions, and identification marking. Examination shall be limited to the examinations that may be performed without disassembling the units. Examination shall also include a check of all adjustments (see 3.3 through 3.4.11 and 3.6 through 3.9).

4.7 Methods of inspection. Unless otherwise specified herein, the FOTS shall be fully assembled and energized throughout the duration of each test procedure.

4.7.1 Accuracy. The sensitive portion of the sensor head shall be immersed in a temperature bath and the FOTS output signal shall be measured at 5, 25, 50, 75, and 95 percent of the FOTS span. Accuracy shall conform to 3.5.1.

4.7.2 Repeatability. The procedure specified in 4.7.1 shall be repeated three successive times while maintaining temperature bath conditions. The FOTS sensor head shall be removed from the bath for not less than two minutes and reimmersed at a repeatable depth for consecutive measurements at each of the five temperature points. FOTS output signal shall conform to 3.5.2.

4.7.3 Reference measurement. The sensitive portion of the sensor head shall be immersed in a temperature bath and the FOTS output signal shall be measured at 5, 50, and 95 percent of the FOTS span. System accuracy shall conform to 3.5.3.

4.7.4 Sensitivity factor. The sensitivity factor shall be determined using the following procedure. Immerse the sensitive portion of the sensor head in a temperature bath and allow ample time to stabilize at a temperature of 80 ± 5 percent of span. Measure both the bath temperature and the FOTS output signal. Increase the bath temperature by an amount not greater than 1 percent of the FOTS span. Measure both the new bath temperature and the FOTS output

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signal. Calculate the change in both bath temperature and FOTS output signal as a percentage of the FOTS span. Determine the ratio of the output signal percentage change to applied bath temperature percentage change in terms of FOTS span. Repeat this procedure for a bath temperature decrease not greater than 1 percent of span. Performance shall conform to 3.5.4.

4.7.5 Response time. The response of the FOTS shall be determined using the following procedure:

(a) Bath medium: water.
(b) Origin bath temperature: 10°C.
(c) Destination bath temperature: 82°C.
(d) Initial temperature (T1): 26°C.
(e) Final temperature (T2): 62°C.

The response time shall be determined as the time required for the FOTS output signal to track from T1 to T2 after the sensor head is transferred from the origin bath to the destination bath. The response time of the FOTS shall be the average of not less than six readings taken during not less than six consecutive trials. Performance shall conform to 3.5.5.

4.7.6 Warm-up time. Warm-up time shall be determined using the following procedure. Allow the sensitive portion of the sensor head to stabilize at a bath temperature of 80 ± 5 percent of the FOTS span. Deenergize the FOTS for not less than 2 hours. Energize the FOTS and monitor the output signal as necessary to ensure conformance to 3.5.6.

4.7.7 Steady-state supply voltage and frequency (ac) or supply voltage (dc). The FOTS output signal shall be monitored for not less than 15 minutes at normal, maximum, and minimum steady-state voltage and frequency conditions as specified in 3.4.6. These conditions are labeled A through E (ac input power) and F through H (dc input power) in table VII. The FOTS shall be subjected to each of the applicable power supply conditions for not less than one hour at ambient temperatures of 0, 25 and 65°C. The FOTS signal output shall be measured after not less than one hour at each test condition. This test may be performed in conjunction with the temperature test (see 4.7.13). Performance shall conform to 3.5.7.

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TABLE VII. Steady-state supply voltage and frequency

Condition	Voltage			Frequency		
	Lower Limit	Normal	Upper Limit	Lower Limit	Normal	Upper Limit
(for ac)						
A		115			60	
B	107			58		
C			123	58		
D			123			62
E	107					62
(for dc)						
F	23.5			N/A	N/A	N/A
G		28		N/A	N/A	N/A
H			32.5	N/A	N/A	N/A

4.7.8 Transient supply voltage and frequency (ac) or supply voltage (dc). The FOTS output signal shall be monitored at ambient temperature during each test procedure defined below. Performance shall conform to 3.5.8.

4.7.8.1 Transient voltage upper limit (ac). With the FOTS operating at a voltage of 123 Vac, the voltage shall be increased to 138 Vac, and then decreased back to the steady-state voltage of 123 Vac in a 2 second period.

4.7.8.2 Transient voltage lower limit (ac). With the FOTS operating at a voltage of 107 Vac, the voltage shall be decreased to 92 Vac, and then increased back to the steady-state voltage of 107 Vac in a 2 second period.

4.7.8.3 Transient frequency upper limit (ac). With the FOTS operating at a frequency of 62 Hz, the frequency shall be increased to 63.5 Hz, and then decreased back to the steady-state frequency of 62 Hz in a 2 second period.

4.7.8.4 Transient frequency lower limit (ac). With the FOTS operating at a frequency of 58 Hz, the frequency shall be increased to 56.5 Hz, and then increased back to the steady-state frequency of 58 Hz in a 2 second period.

4.7.8.5 Transient voltage upper limit (dc). With the FOTS operating at a voltage of 32.5 Vdc, the voltage shall be increased to 34.5 Vdc, and then decreased back to the steady-state voltage of 32.5 Vdc in a 2 second period.

4.7.8.6 Transient voltage lower limit (dc). With the FOTS operating at a voltage of 23.5 Vdc, the voltage shall be decreased to 21.5 Vdc, and then increased back to the steady-state voltage of 23.5 Vdc in a 2 second period.

4.7.9 Insulation resistance. The insulation resistance of the opto-electronics module circuits shall be determined by applying 50 Vdc between electrical input and output circuits and between these circuits and

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ground. The insulation resistance measurement shall be made immediately after a 2 minute period of uninterrupted test voltage application. If the indication of insulation resistance meets the specified limit (see 3.5.9) and is steady or increasing, the test may be terminated before the end of the 2 minute period. Performance shall conform to 3.5.9.

4.7.10 Power interruption. The FOTS shall be energized and the output signal monitored at ambient temperature throughout the test. The external power supply shall be suddenly interrupted, and after an interval not less than 4 seconds, the power supply shall be reapplied to the FOTS. Performance shall conform to 3.5.10.

4.7.11 Short circuit. The FOTS shall be deenergized and the positive and negative electrical output terminals of opto-electronics module shall be connected directly together, with no load resistance. The FOTS shall be energized for five minutes, then deenergized and the short circuit removed. The FOTS shall be energized and the output signal measured at ambient temperature. Performance shall conform to 3.5.11.

4.7.12 Line voltage reversal (dc). The positive 28 Vdc power input shall be applied to connector pin "B" and the reference shall be applied to connector pin "A". The power supply shall be energized for a period of not less than 10 minutes. The FOTS output signal shall be measured at ambient temperature. The FOTS shall be deenergized and the power supply connections shall be reversed to normal polarity. The FOTS shall then be energized for not less than 5 minutes before the output signal is measured at ambient temperature. Performance shall conform to 3.5.12.

4.7.13 Temperature. The FOTS shall be positioned in an environmental chamber in an energized state, and shall be subjected to the following conditions consecutively:

(a) Hold temperature at 0 ± 2°C for 24 hours.
(b) Increase temperature in steps of 10°C at 30 minutes per step until 65 ± 2°C is achieved.
(c) Hold temperature at 65 ± 2° for 24 hours.
(d) Reduce temperature in steps of 10°C at 30 minutes per step until 25 ± 2°C is achieved.

Measure the FOTS output signal at one point during the last hour of operation at each temperature plateau (that is 0, 25, and 65°C). Performance shall conform to 3.5.13.

4.7.14 Humidity. The FOTS shall be subjected to the following conditioning and test. The FOTS shall be energized throughout the entire procedure. Performance shall conform to 3.5.14.

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4.7.14.1 Conditioning. In order to establish a reference condition for the measurement of operating parameters and a valid basis for comparison of the effects of the conditioning to follow, the complete equipment shall be dried at a temperature not less than 40°C or more than 50°C for at least 2 hours.

4.7.14.2 Reference measurement. Following the conditioning specified in 4.7.14.1, the FOTS signal output shall be measured at 25 ± 5°C and 50 ± 5 percent relative humidity. Performance shall conform to 3.5.14.

4.7.14.3 Temperature cycling. The FOTS shall then be subjected to four 24-hour cycles of temperature variation consisting of 18 hours at 65 ± 5°C and 6 hours at 25 ± 5°C. The relative humidity shall be maintained above 95 percent during the steady-state conditions. The transitions between temperatures shall be accomplished within the 6 hour period so that the time at the high temperature is not less than 18 hours. Each transition shall be not greater than 1 hour if the sensor remains in the chamber, or 15 minutes if a two chamber method is employed. The relative humidity need not be controlled during the transition period.

4.7.14.4 Measurements during temperature cycling. During the second cycle, the measurement required in 4.7.14.2 shall be performed at the end of the high temperature period (65 ± 5°C) with the sensor remaining in the chamber, before the temperature is decreased to 25°C.

4.7.14.5 Measurements after temperature cycling. After the four complete cycles, the measurement required in 4.7.14.2 shall be performed at 25 ± 5°C with the sensor remaining in the chamber. In addition, the FOTS components shall be examined for water penetration.

4.7.15 Enclosure. The FOTS sensor head and not less than a two meter section of the attached cable interconnect shall be mounted to a surface which extends not less than 1 meter beyond the sensor head on all sides so that splashing shall be produced by directing the water stream on that surface. The FOTS shall be deenergized throughout the test. The water stream shall be a coarse spray with a flow rate of not less than 57 liters per second and a head pressure of not less than 3 meters. A head pressure of 3 meters is defined as sufficient water pressure so that if directed straight up, the stream of water shall rise to a height of not less than 3 meters. The distance from the nozzle to the enclosure under test shall be approximately 2 meters. The time of the test shall be not less than 5 minutes with approximately equal portions of time for spray on each surface, including joints of the enclosure and at the surface at which the enclosed equipment is mounted. The FOTS components shall be examined at the immediate conclusion of the test for moisture penetration. The FOTS shall be energized and the output signal monitored at ambient temperature at the conclusion the test. Performance shall conform to 3.5.15.

4.7.16 Salt spray. The FOTS sensor head and not less than a 1 meter section of the attached cable interconnect shall be subjected to salt spray in accordance with method 101 of MIL-STD-202. This test is not applicable to the

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embedded sensor configuration since it will be secured in a bearing. The FOTS shall be deenergized throughout the test. Duration of the test shall be 96 hours. The FOTS components shall be examined at the immediate conclusion of the test for corrosion and moisture penetration. The FOTS shall be energized and the output signal monitored at ambient temperature at the conclusion of the test. Performance shall conform to 3.5.16.

4.7.17 Overtemperature. The sensitive portion of the FOTS sensor head shall be rapidly transferred from ambient conditions to a temperature bath stabilized at 125 percent of the FOTS full scale temperature. The sensor head shall remain immersed in the bath for 15 minutes and then returned to ambient conditions. A reference measurement (see 4.7.3) shall be performed at the conclusion of the overtemperature test. Performance shall conform to 3.5.17.

4.7.18 Accelerated life. The sensitive portion of the sensor head shall be thermally cycled between the following temperatures for 1500 cycles at a rate not greater than 2 cycles per minute:

Sensor range	Temperature cycling points
-40 to 150°C	Ambient to -30 ± 2°C
0 to 300°C	Ambient to 285 ± 2°C

The FOTS output signal shall be monitored during thermal cycling and observed for operation without interruption. A reference measurement (see 4.7.3) shall be performed at the conclusion of the thermal cycling test. Performance shall conform to 3.5.18.

4.7.19 Vibration. The FOTS shall be tested in accordance with MIL-STD-167-1, type I. Test fixtures shall be designed to simulate the shipboard installation of the sensor type. The FOTS output signal shall be monitored throughout the test and visually examined for operation without interruption, especially at resonant frequencies. The FOTS shall be visually examined for physical damage during and at the conclusion of the test. A reference measurement in accordance with 4.7.3 shall be performed at the conclusion of the test. Performance shall conform to 3.5.19.

4.7.20 Shock. The FOTS shall be fully assembled, energized, and tested in accordance with grade A, class I, type A of MIL-S-901. Test fixtures shall be designed to simulate the shipboard installation of the sensor type. The FOTS shall be visually examined for physical damage after each hammer blow and at the conclusion of the test. A reference measurement in accordance with 4.7.3 shall be performed at the conclusion of the test. Performance shall conform to 3.5.20.

4.7.21 EMI emission and susceptibility. EMI tests shall be in accordance with the following test methods specified in MIL-STD-462: CE101, CE102, CS101, CS114, CS116, RE101, RE102, RS101, RS103 and RS105. Performance shall conform to 3.5.21.

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4.7.22 Pressure. A reference measurement (see 4.7.3) shall be performed prior to the test to ensure the sensor head is not defective. The sensor head under test shall be installed in a fixture made from mild steel. The critical dimensions to which the fixture shall conform are shown on figure 5 and in table VIII. The 19.05 mm (0.75 inch) diameter counterbore shall contain ASTM B 23 (grade 2) babbitt flush with the test fixture surface. The babbitt surface shall then be exposed to a gauge pressure of 123.26 ± 0.07 MPa (1800 ± 10 pounds per square inch) for not less than 5 minutes. The FOTS output signal shall be monitored during the test and examined for operation without interruption. The babbitt shall be removed from the 19.05 mm (0.75 inch) counterbore and a reference measurement (see 4.7.3) shall be performed. Performance shall conform to 3.5.22.

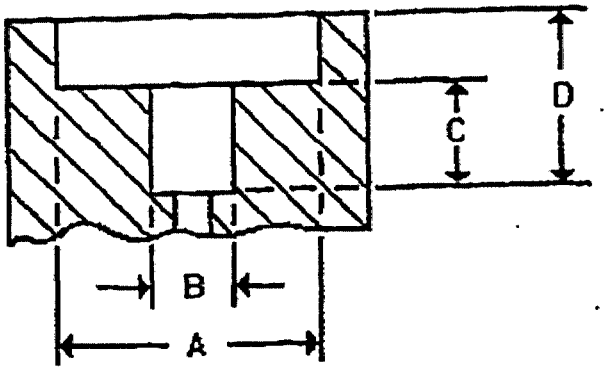


FIGURE 5. Pressure test fixture.

TABLE VIII. Pressure test fixture critical dimensions.

Pressure test fixture dimensions in mm(in.); See figure 5	
A	19.05 ± 0.4 Dia. (0.75 ± 0.016)
B	7.06 +0.08/-0.00 Dia. (0.278 +0.003/-0.00)
C	6.35 +0.25/-0.00 (0.25 +0.01/-0.00)
D	11.11 ± 0.4 (0.438 ±0.016)

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4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance MIL-E-17555.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7).

5.1 Packaging requirements. FOTS shall be individually preserved level A, C, or commercial, packed level A, B, C or commercial as specified (see 6.2) and marked in accordance with MIL-E-17555 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). Unless otherwise specified (see 6.2), method IV shall apply for level A preservation.

5.2 Marking. In addition to any special marking required (see 6.2), unit and intermediate packs and shipping containers and palletized unit loads shall be marked in accordance with ASTM-D-3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. FOTS are intended to monitor temperatures in shipboard engineering plants through interfacing with readout devices.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of FOTS required.
(c) FOTS classification required (see 1.2).
(d) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) When the first article inspection is required (see 3.2, 4.4 and 6.5).
(f) Digital optical output requirements (see 3.4.7.3).
(g) Requirements including data format for digital electrical output (see 3.4.7.4).
(h) The connector assembly when digital optical or digital electrical output is required (see 3.4.8).
(i) Opto-electronics module (type B or C) requirements (see 3.4.2.2 and 3.4.2.3).
(j) Cable length required (see 3.4.3.1 and 3.4.3.2).
(k) End-use for sensors if other than surface ships (see 3.5.21).

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(l) If responsibility for inspection equipment shall be other than specified (see 4.1).
(m) Final disposition of first article test samples (see 4.4.4).
(n) Levels of preservation and packing required (see 5.1).
(o) Applicable acquisition requirements of MIL-E-17555 (see 5.1).
(p) When other than preservation method IV applies to level a preservation (see 5.1)
(q) If special marking is required (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475.1 exempts the requirement for a DD Form 1423.

Paragraph No.	DID Number	DID Title	Options
Appendix	DI-DRPR-80651	Engineering Drawings	Level 3
4.4	DI-NDTI-80809	Tests/Inspection Reports	____

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DOD 5010.12-L AMSDL must be listed on a separate CDRL (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the contracting officer should specify that the items should be first article samples. The first article samples shall consist of the units specified in 4.3.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article units. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Subcontracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Subject term (key word) listing.

Bare-bulb
Embedded
Single Channel
Thermowell
Sensor head
Opto-electronics module

Preparing activity:
Navy - SH
(Project 6040-N002-02)

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ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix contains the format and content preparation instructions for the development of and revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction, and principle of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is applicable only when data item description DI-DRPR-80651 is cited on the DD Form 1423. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY
MIL-I-45208 - Inspection System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

20.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y14.1 - Drawing Sheet Size and Format. (DOD adopted)
200 - Standard Reference Designations for Electrical and Electronics Parts and Equipments. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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30. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

30.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

30.2 Content and format. One conformance verification drawing shall be developed for each specific type of instrumentation. It shall include all ranges, sizes, connections and other variations. The conformance verification drawing shall include the following minimum information (except as specified in 10.1) and shall be developed to the format in 30.3 through 30.9. A symbol list shall accompany all drawings and schematics.

30.3 Descriptive data.

(a) Instrument identification numbering system for instrumentation. This numbering system shall include, but may not be restricted to, the classification variables.
(b) Instrumentation identification number system for replaceable parts.
(c) Size, operating data, ranges, scale markings, and other data for proper selection.
(d) Test approval data, presenting the following information in tabular form:
    (1) Specification classification (unique alphanumeric variables to designate instruments).
    (2) Test report number and date.
    (3) Facility where test was conducted.
    (4) Authorized Government activity (NAVSEA) approval letter and date.
(e) A statement that instrumentation is in accordance with the requirements of the applicable instrumentation specification or acquisition document and to referenced specifications.
(f) Conformance verification drawing acceptance data, presenting the following information in tabular form:
    (1) Authorized Government activity (NAVSEA) acceptance letter and date.
    (2) Revision number.

30.4 Details of construction.

(a) Two or more representative assembly views, as required, to show clearly the details of the design, construction, and assembly of the instrumentation and to identify each part and its location. Identification of parts shall correspond to the list of materials. Assembly shall show how all mechanical parts are joined or attached.
(b) Sectional views or notes as necessary to show internal details.
(c) Details such as entrance provisions, gaskets, fastening techniques, welding symbols, mounting requirements, and other details as applicable.

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(d) Module enclosure requirements for separately mounted modules.
(e) Dimensions required to ensure interchangeability.
(f) Schematic of indicating system.
(g) Description of the nature and purpose of any adjustments.
(h) Welding procedures with acceptance data, including acceptance letter, report numbers and dates.
(i) Any special features.
(j) Complete weight.
(k) Location, size, and type of connections.
(l) Identification of quality control documents which show conformance with MIL-I-45208 or the quality control specification that is listed in the applicable instrumentation specification or acquisition document. The acceptance letter shall be referenced.

30.5 Parts list. The following information shall be presented in tabular form:

(a) Item number (corresponding to flag number identifying the part on the conformance verification drawing).
(b) Quantity of each part required per assembly.
(c) Name of part with sufficient information to readily identify the part (for example, screws: thread size, length and type of head, shall be specified).
(d) Material of part.
(e) Material specification (military, federal, or non-Government document number or Government activity (NAVSEA) drawing number).

NOTE: When substitution of a material specification is made, it is the responsibility of the contractor to provide written documentation to substantiate that the substituted material is equivalent to the specified material.

(f) Type, class, grade, size, military designation, or other classification of any referenced specification.
(g) Part number or identification assigned by assembly supplier.
(h) Name of actual manufacturer of part (when applicable).
(i) Part number or identification assigned by part supplier (when applicable).
(j) Onboard repair parts. Parts that are appropriate for, or are supplied as, onboard repair parts shall be indicated by a symbol in this column.
(k) Remarks column. Finishes, platings, or coatings along with the applicable specification or other requirements should be specified in this column.

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30.6 Table for special tools. Special tools required for the instrumentation shall be presented in the following tabular form:

(a) Item number (corresponding to flag number identifying the tool on the conformance verification drawing).
(b) Quantity of each tool required per instrument.
(c) Name (description) of tool (include generic name).
(d) Tool specification (military, federal, or non-Government document number or Government activity (NAVSEA) drawing number).
(e) Tool number or identification assigned by assembly supplier.
(f) Name of actual manufacturer of tool (when applicable).
(g) Tool number or identification assigned by tool supplier (when applicable).
(h) Description of tool's application.
(i) Remarks column. Special techniques or other usage requirements shall be explained in this column.

30.7 Electrical system schematics. Electrical schematics shall be included in the verification conformance drawing. A complete schematic shall contain all the parts in the electrical systems. When confusion over its function could result, a simplified schematic containing only the major components shall also be provided. This simplified schematic may be presented in block diagram format.

(a) Complete electrical system schematic. A single schematic shall represent clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic shall contain all parts (including components, connectors, alarms, and so forth) which make up the electrical system. The following features shall be incorporated into the schematic.
    (1) In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.
    (2) A thin, broken line shall be used to represent the boundaries of each unit or subassembly. Terminals, to which external connections are made, shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.
    (3) Each part (such as resistors, capacitors, relays, and so forth) shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI 200) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of electrical current or signal flow through the circuit.
    (4) In addition to where the reference designation, parts not conforming to a military specification, where permitted, shall have the following information noted adjacent to the part.

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Where numerical values are given, a code shall be noted to designate the units used:

a. Resistors - resistance, power rating, and tolerances. If variable, an arrow to indicate clockwise rotation of the control shaft.
b. Capacitors - capacitance, voltage rating, and tolerance.
c. Reactors - inductance and voltage rating.
d. Semi-conductors - type number (JEDDC number is adequate).
e. Integrated circuits - operational symbol diagram of input-output relationship, terminal numbering corresponding to a representative schematic, and type number.

(5) Supply voltages, phases, and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.
(6) Table for troubleshooting of electrical system. The content table shall include each test point, as identified on the electrical system schematic, with voltage, waveform or other electrical parameter that should be measured at each test point.

30.8 Selection and installation considerations.

(a) Performance data.
(1) Accuracy (in percent of span).
(2) Shock and vibration classification.
(3) Degree of water tightness of the enclosure.
(4) Electromagnetic interference and pulse susceptibility.
(b) Dimensional outline of the instrumentation showing overall and principle dimensions in sufficient detail to establish space requirements in all directions necessary for installation, servicing, exclusive of space required for operator observation of the indication.
(c) Special considerations which may affect selection or installation.
(1) Ambient temperature range.
(2) Calibration points and adjustments.
(3) Orientation.
(4) Location of instrumentation relative to vibrating equipment.
(5) Protection of the instrumentation from pulsations and spikes in the parameter being measured.
(6) Selection of the instrumentation range relative to the operating range of the system.
(7) Application for each type connection.
(8) Cleaning procedure or reference to the cleaning procedure used.
(9) Selection of the instrumentation for compatibility (materials, temperature, pressure, and so forth) with the ambient environment and with the parameter being measured.

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30.9 Drawing format.

(a) Unless otherwise approved by the authorized Government activity, a maximum of three sheets shall be allotted for single functioned system instrumentation and a maximum of ten sheets for instrumentation containing a multiple functioned system. A single functioned system is one that performs only one operation such as: expands or contracts a pressure elastic element, steps up or steps down the voltage, conditions one electrical signal, or winds or unwinds a bimetallic element. A multiple functioned system is one that contains two or more single functioned systems.
(b) Each sheet shall be zoned.
(c) Title block shall be included on each drawing sheet and shall include the following information:
  (1) Title, drawing number, and revision letter. Each sheet shall contain the same title, drawing number, and revision letter.
    a. Title - the title shall consist of the name by which the instrumentation is known.
    b. Drawing number - the drawing number shall consist of alphanumeric characters which may be separated by dashes or slashes. The total number of characters in the drawing number (including dashes and slashes) shall be not greater than 15. Blank spaces are not permitted within the drawing number.
    c. Revision letter - the revision letter shall denote the latest approved version of the drawing. The revision letter of conformance verification drawing shall not be changed until all the changes under that revision have been accepted in writing by the authorized Government activity (NAVSEA). No changes made to the conformance verification drawing shall be considered a revision until after the initial version of the conformance verification drawing has been accepted in writing by the authorized Government activity. After initial submittal of the conformance verification drawing, no changes shall be made during the initial drawing review process unless the change is either requested by the authorized Government activity or the change is documented in written correspondence by the contractor.
  (2) Sheet _____ of _____.
  (3) Tolerance on dimensions for fractions, decimals, and angles. Units of the dimensions specified on the conformance verification drawing.
  (4) Contractor acceptance block (appropriate signatures and dates).
  (5) Commercial and Government entity (CAGE) code for manufacturer.
  (6) Scale.
  (7) Reference drawings.
  (8) Manufacturer's name and address.
  (9) Drawing size.

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(d) A revision block shall be included on each sheet of the conformance verification drawing and shall contain the following information in tabular form:
(1) Revision letter.
(2) Description of revision.
(3) Acceptance letter serial number and originator identification.
(4) Acceptance data.
(e) Sheet size and format not specified herein shall be in accordance with ANSI Y14.1.
(f) Classification designations. No Government security classification designation such as confidential or secret shall appear on the conformance verification drawing unless a particular classification is specified by the Government.

40. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE

40.1 Acceptance. Acceptance shall be granted by the authorized Government activity only after the conformance verification drawing is found to meet all the requirements specified in 20. through 40.1.

50. INSTRUCTIONS

50.1 Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA shall be designated as the activity that accepts or disapproves the conformance verification drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-24795(SH)	2. DOCUMENT DATE (YYMMDD) 95/06/22

3. DOCUMENT TITLE
SENSORS, TEMPERATURE, FIBER OPTIC (NAVAL SHIPBOARD USE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY TECHNICAL POINT OF CONTACT (TOPC):

a. NAME	b. TELEPHONE (Include Area Code)
Mr. Frank LeLand SEA 03J4 ALL CORRESPONDENCE SHOULD BE ADDRESSED TO:	(1) Commercial (703) 602-6137 (2) AUTOVON 332-6137
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 03R42, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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